By property (4):

$$E[2\beta'(F-\overline{F})\varepsilon]=0$$

By property (2) and property (7) we have,

$$E[\varepsilon'\varepsilon]=\begin{bmatrix}\sigma_{e1}^2 & 0 & \cdots & 0\\ 0 & \sigma_{e2}^2 & \cdots & 0\\ \vdots & \vdots & \ddots & \vdots\\ 0 & 0 & \cdots & \sigma_{en}^2\end{bmatrix}=\Lambda$$

which is the idiosyncratic variance matrix and is a diagonal matrix consisting of the variance of the regression term for each stock.

By property (3) and property (5) the factor covariance matrix is:

$$E[(F-\overline{F})^2]=\begin{bmatrix}\sigma_{f1}^2 & 0 & \cdots & 0\\ 0 & \sigma_{f2}^2 & \cdots & 0\\ \vdots & \vdots & \ddots & \vdots\\ 0 & 0 & \cdots & \sigma_{fk}^2\end{bmatrix}=\mathrm{cov}(F)$$

The factor covariance matrix will be a diagonal matrix of factor variances. In certain situations there may be some correlation across factors. When this occurs, the off-diagonal entries will be the covariance between the factors. Additionally, the beta sensitivities may be suspect meaning that we may not know the true exposures to each factor and we will have some difficulty determining how much that particular factor contributes to returns. However, the covariance calculation will be correct providing we include the true factor covariance matrix.

Finally, we have our covariance matrix derived from the factor model to be:

Covariance Matrix

$$C=\beta'\mathrm{cov}(F)\beta+\Lambda \tag{6.36}$$

This matrix can be decomposed into the systematic and idiosyncratic components. Systematic risk component refers to the risk and returns that are explained by the factors. It is also commonly called market risk or factor risk. The idiosyncratic risk component refers to the risk and returns that are not explained by the factors. This component are also commonly called stock specific risk, company specific, and diversifiable risk. This is shown as:

$$C=\underbrace{\beta'\mathrm{cov}[F]\beta}_{\substack{\text{Systematic}\\ \text{Risk}}}+\underbrace{\Lambda}_{\substack{\text{Idiosyncratic}\\ \text{Risk}}} \tag{6.37}$$